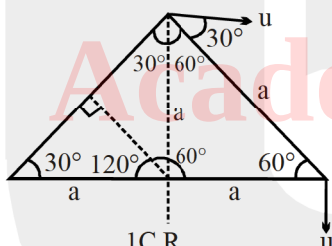
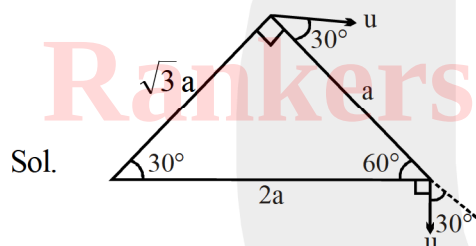


HINTS AND SOLUTIONS

- Q.1 C
Q.2 C
Q.3 B
Q.4 C
Q.5 A
Q.6 B
Q.7 B
Q.8 C
Q.9 B
Q.10 B
Q.11 C
Q.12 C
Q.13 C
Q.14 A
Q.15 A
Q.16 B
Q.17 BCD



$$V_c = w_a = V$$

Q.18 ABC

Sol. C has max torque $\Rightarrow \omega_C$ max
B - energy dissipated

$$mg \sin \theta - \mu mg \cos \theta = ma_B \Rightarrow a_B > \frac{2}{3} g \sin \theta$$

$$a_C = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta$$

Q.19 ACD

Q.20 AC

Sol. $N_A + f_B = 2mg$

$$N_B = f_A$$

$$mgR + -f_A R - f_B R = 0$$

$$mg = f_A + f_B$$

Q.21 Ans. (A)-S (B)-R (C)-Q (D)-R

Sol._{SJ} In (A) we try to decrease ω of the rear wheel then friction will try to increase it. On front wheel at the point of contact with the ground $\omega \times r$ is greater than v as v decreases due to break. So that friction will act in forward direction on it.

In (B) we try to decrease ω of the front wheel friction will try to increase it.

In (C) sliding will start hence f_k will act.

In (D) we try to increase ω of rear wheel then friction will try to decrease it.

Q.22 Sol. Conserving angular momentum about hole

$$mvr_0 = mv_1 r_0/2$$

$$v_1 = 2v$$

$$\text{Work done} = \Delta KE = (1/2)m[(2v)^2 - v^2] = (3/2)mv^2$$

Q.23 Ans. 0009

Sol. $\tau - fR = 0.45 \alpha$

$$f = 10a = 10 \times 0.2 = 20\text{N}$$

$$\tau - 20 \times 0.3 = 0.45 \times \frac{0.2}{0.3} = 3$$

$$\tau = 9 \text{ N-m}$$

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